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Python Programming

PY101 | Suggested sequence · Semester 1 | Student-authored study guide

How to use these notes

Beginner Python: values, control flow, reusable functions, collections, files, and testing. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Values, expressions, and decisions

Names refer to objects. Use descriptive identifiers, convert input explicitly, and distinguish equality (==) from assignment (=). Boolean expressions can be combined with and/or/not; if/elif/else selects a branch.

Worked example

```
score = float(input("Score: ")); if 0 <= score <= 100: print("valid") else: print("check
range")
```

Check your understanding

- What type does input() return?
- Why validate a number before calculating with it?

MODULE 02 · Loops and collections

Lists preserve order, tuples represent fixed sequences, sets hold unique values, and dictionaries map keys to values. for loops iterate over items; while loops repeat while a condition remains true. Prefer clear comprehensions for simple transformations.

Worked example

```
counts = {}; for word in ["data", "science", "data"]: counts[word] = counts.get(word, 0) + 1
```

Check your understanding

- When is a dictionary preferable to parallel lists?
- How can an infinite while loop be prevented?

MODULE 03 · Functions, files, and tests

Functions isolate a task and make assumptions explicit. Return results rather than printing inside reusable logic. Use with open(...) to close files reliably, handle only expected exceptions, and test normal, boundary, and invalid cases.

Worked example

```
def mean(values):
    if not values: raise ValueError("values cannot be empty")
    return sum(values) / len(values)
```

Check your understanding

- What is the difference between return and print?
- Name three useful test cases for a function.

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.